

What is biology?

THE SCIENCE THAT INVESTIGATES EVERY FORM OF LIFE—HOW IT SURVIVES AND WHERE IT ORIGINATED.

Biology, or life science, is a vast subject that studies life at all scales, from the inner workings of a microscopic cell to the way whole forests behave.

What is life?

All life shares seven basic characteristics. These are not exclusive to life, but only living things have all seven. For example, a car can move, it “feeds” on fuel, excretes exhaust, and may even sense its surroundings, but these four characteristics do not make the car alive.

▷ The seven characteristics

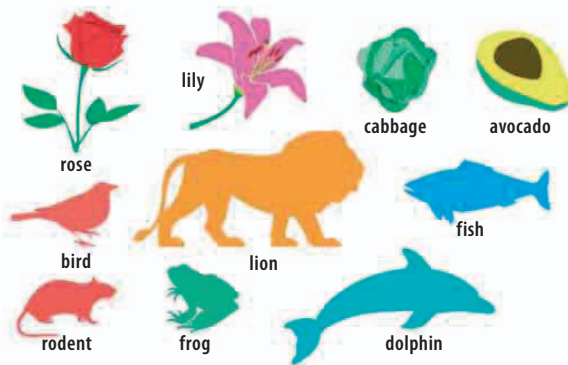
Living things, or organisms, are incredibly varied. Even so, they all share the same seven characteristics that set them apart from nonliving things.

THE SEVEN REQUIREMENTS FOR LIFE

Requirement	Description
movement	altering parts of its body in response to the environment
reproduction	being able to make copies of itself
sensitivity	able to sense changes in the surroundings
growth	increasing in size for at least a period of its life
respiration	converting fuels (eg, food) into useful energy
excretion	removing waste materials from its body
nutrition	acquiring fuel to power and grow its body

Taxonomy

The field of biology that organizes, or classifies, organisms is called taxonomy. Modern taxonomy groups organisms according to how they are related to each other (rather than just how they look). It involves placing all organisms in groups, or taxons, arranged in this hierarchy: domain, kingdom, phylum (or division in the plant kingdom), class, order, family, genus, and species. Animals and plants are part of the largest domain, Eukaryota.

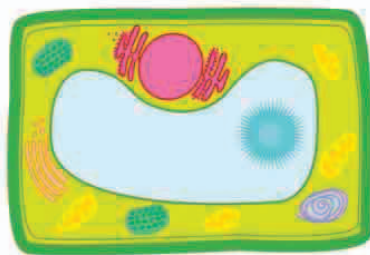


◁ Classification

Taxonomy (see pages 20–21) shows us that some of these organisms are more closely related than others. For example, animals belong to the animal kingdom, whereas plants belong to the plant kingdom.

Microbiology

A cell is the smallest unit of life and that is what microbiologists study. They use microscopes to see inside cells and investigate how their minute inner machinery, often called organelles, functions to keep the cells alive. Microbiology has shown that not all cells are the same, which helps explain how bodies work and gives clues to how life started and has since evolved.



plant cell

◁ Seeing in detail

This cutaway artwork shows the inner structures of a plant cell. Microbiologists (see page 23) view the finest details using powerful electron microscopes, which use a beam of electrons instead of light to magnify cells.